ACTIVITIES & PROGRAMMES

Antimicrobial resistance – reducing environmental contamination

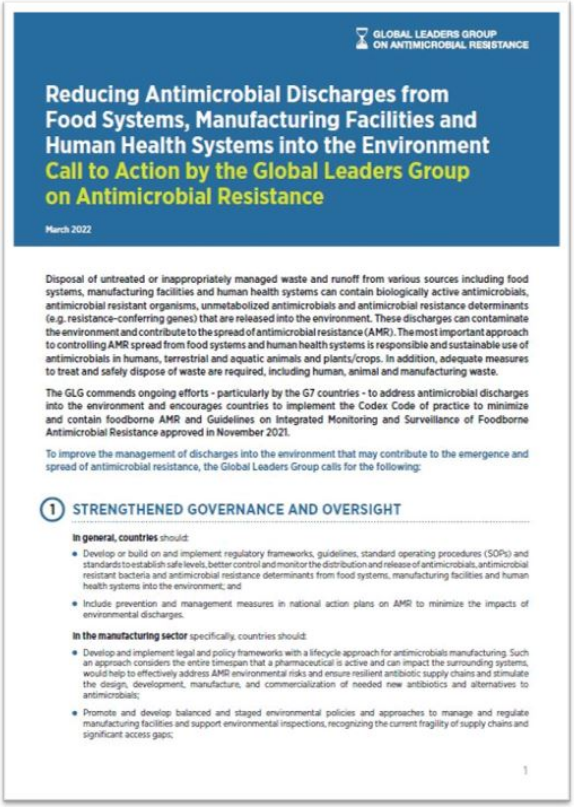
KEYWORDS

#antimicrobial resistance (AMR), #environmental impact, #Interagency Coordination Group (IACG), #World Organisation for Animal Health (WOAH).

All pollution ends up in the marine environment. © Ólafur Valsson

The Global Leaders Group (GLG) on Antimicrobial Resistance (AMR) *was established in November2020 following a recommendation of the Interagency Coordination Group on AMR to strengthen global political momentum and leadership on AMR[1]. The inaugural meeting of the GLG took place in January2021.*

The GLG consists of world leaders and experts from across sectors working together to accelerate political action on AMR. The mission of the GLG on AMR is to collaborate globally with governments, agencies, civil society and the private sector through a One Health approach, and to advise on and advocate for prioritised political actions for the mitigation of drug-resistant infections through responsible and sustainable access to, and use of, antimicrobials.



GLOBAL LEADERS GROUP ON ANTIMICROBIAL RESISTANCE

Reducing Antimicrobial Discharges from Food Systems, Manufacturing Facilities and Human Health Systems into the Environment
Call to Action by the Global Leaders Group on Antimicrobial Resistance

March 2022

Disposal of untreated or inappropriately managed waste and runoff from various sources including food systems, manufacturing facilities and human health systems can contain biologically active antimicrobials, antimicrobial resistant organisms, unmetabolized antimicrobials and antimicrobial resistance determinants (e.g. resistance-conferring genes) that are released into the environment. These discharges can contaminate the environment and contribute to the spread of antimicrobial resistance (AMR). The most important approach to controlling AMR spread from food systems and human health systems is responsible and sustainable use of antimicrobials in humans, terrestrial and aquatic animals and plants/crops. In addition, adequate measures to treat and safely dispose of waste are required, including human, animal and manufacturing waste.

The GLG commends ongoing efforts - particularly by the G7 countries - to address antimicrobial discharges into the environment and encourages countries to implement the Codex Code of practice to minimize and contain foodborne AMR and Guidelines on Integrated Monitoring and Surveillance of Foodborne Antimicrobial Resistance approved in November 2021.

To improve the management of discharges into the environment that may contribute to the emergence and spread of antimicrobial resistance, the Global Leaders Group calls for the following:

1 STRENGTHENED GOVERNANCE AND OVERSIGHT

In general, countries should:

- Develop or build on and implement regulatory frameworks, guidelines, standard operating procedures (SOPs) and standards to establish safe levels, better control and monitor the distribution and release of antimicrobials, antimicrobial resistant bacteria and antimicrobial resistance determinants from food systems, manufacturing facilities and human health systems into the environment; and
- Include prevention and management measures in national action plans on AMR to minimize the impacts of environmental discharges.

In the manufacturing sector specifically, countries should:

- Develop and implement legal and policy frameworks with a lifecycle approach for antimicrobials manufacturing. Such an approach considers the entire timespan that a pharmaceutical is active and can impact the surrounding systems, would help to effectively address AMR environmental risks and ensure resilient antibiotic supply chains and stimulate the design, development, manufacture, and commercialization of needed new antibiotics and alternatives to antimicrobials;
- Promote and develop balanced and staged environmental policies and approaches to manage and regulate manufacturing facilities and support environmental inspections, recognizing the current fragility of supply chains and significant access gaps;

1

The GLG on AMR has, since its inauguration, issued several information notes, statements and calls to action, including a call to action on reducing antimicrobial discharges from food systems, manufacturing facilities and human health systems into the environment[2].

This call to action on discharges into the environment addresses the disposal of untreated or inappropriately managed waste and runoff from different sources such as human health systems, food systems and manufacturing facilities, because these can contain biologically active antimicrobials, antimicrobial-resistant organisms, unmetabolised antimicrobials and antimicrobial resistance determinants (e.g.resistance-conferring genes) that contaminate theenvironment.

The call to action is timely, as evidence builds on how this contamination of the environment can contribute to the spread of AMR. It highlights the necessity for strengthened governance and oversight, a fundamental requirement of activities to mitigateAMR. It points out the importance of *containing at source* and improving discharge management. Preventing environmental contamination is pivotal. The call to action emphasises surveillance, innovation and research across public and private sectors into cost-effective and greener waste management technologies.

https://doi.org/10.20506/bull.2022.2.3323

REFERENCES

1. Interagency Coordination Group on Antimicrobial Resistance (IACG) (2019). – No time to wait: Securing the future from drug-resistant infections. Report to the Secretary-General of the United Nations. April 2019.
2. Global Leaders Group on Antimicrobial Resistance (2022). – Reducing antimicrobial discharges from food systems, manufacturing facilities and human health systems into the environment. Call to action by the Global Leaders Group on Antimicrobial Resistance. March 2022.
3. Global Leaders Group on Antimicrobial Resistance (2021). – Antimicrobial resistance and the climate crisis. Information note of the Global Leaders Group on Antimicrobial Resistance. October 2021.
4. Food and Agriculture Organization of the United Nations (FAO), World Organisation for Animal Health (WOAH), World Health Organization (WHO) & United Nations Environment Programme (UNEP) (2021). – Strengthened partnership aims to accelerate coordinated strategy on human, animal and ecosystem health. Joint

FAO/WOAH/WHO/UNEP press release.